

Graphical Abstract

When failing tests are not enough: repair-grounded evaluation of misconception clustering for introductory C programs



Highlights

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- Delta-debugged student repairs yield verified mechanism labels without annotation
- Identical test signatures mix mechanisms: at most 73.9% of real items are placeable
- Output-deviation OAV lifts cluster-mechanism ARI from 0.09 to 0.33 on mutants
- ILA-2 rules over deviation evidence transfer to unseen assignments
- Frozen code embeddings cluster programs by their origin, not by mechanism

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Abstract

Clustering failed programming submissions by the test cases they fail, and explaining each cluster with IF-THEN rules, is a common way to tell instructors which misconceptions a class shares. Whether such clusters group programs by the mechanism of their errors is rarely tested for lack of mechanism labels. We build them without annotation. After replaying 8,607 C90 submissions of a public corpus in an isolated runner (98.42% agreement with the historical grader), we reduce each failing program and the student's next passing program to a verified minimal repair by delta debugging and classify it into twelve program-level mechanisms (339 single-mechanism labels); a second benchmark injects single mechanisms into accepted programs (3,320 mutants). Under a pre-registered protocol, identical test signatures mixed mechanisms: averaged over assignment cohorts, no function of the test-outcome representation could place more than 73.9% of real and 50.7% of injected items. Label-free descriptors of how the output deviates from the oracle raised this ceiling beyond a granularity-matched null and, on controlled data, raised the K-means adjusted Rand index from 0.09 to 0.33; on real repairs the gain (0.29 to 0.35) was not reliable, and the descriptors did not beat simpler output relations. ILA-2 rules over these descriptors transferred to unseen assignments (macro-F1 0.35), whereas a frozen code embedding clustered mutants by source program, not by mechanism. An instructor tool built on these results names a mechanism only when a rule covers most of a group. For mechanism-level feedback, the cheapest improvement is evidence the autograder already has.

Keywords: programming education, misconceptions, clustering, program repair, rule induction, educational data mining

1. Introduction

Instructors of large introductory programming courses cannot read every failed submission, yet they would like to know what the class as a whole gets

wrong so that they can re-teach it. A popular answer is to cluster failed submissions and to summarise each cluster for the instructor. In the most common formulation, each submission is an object described by attribute-value pairs that record which test cases it fails (an object-attribute-value, OAV, representation), the objects are clustered with K-means or a Hamming-distance method, and production rules of the form IF *symptom* THEN *misconception* are induced to explain the clusters. The appeal is clear: test outcomes are available in every autograder, the representation is interpretable, and the rules read like teaching advice.

Whether such clusters group submissions by the *mechanism* of the error is rarely tested, for a simple reason: public datasets of failed programs do not come with mechanism labels, and asking experts to label thousands of programs is expensive and hard to reproduce. Without labels, cluster quality is judged by internal indices such as silhouette, or by the fidelity of surrogate rules to the clusters. Neither says whether two programs that fail the same tests fail them for the same reason.

Figure 1 makes the problem concrete. Of the 89 failing submissions to one assignment of our corpus, 78 fail all four tests, so a test-case OAV puts them in one cluster and any rule that reads it states one conclusion for all of them. The three programs in the figure need three different minimal repairs—a missing newline, a missing zero-padding specifier and a missing pair of parentheses—and their outputs deviate from the expected output in three different ways. What separates them is not the pass/fail vector but *how* each output differs.

This paper builds the missing evaluation data without new annotation and uses it to test the classic pipeline. Our starting point is that a student’s own later submission often repairs the failed one. We replay a public corpus of 8,607 C90 submissions (Orvalho et al., 2024) in an isolated runner, pair each failing submission with the same student’s next passing submission, and use delta debugging to reduce their difference to a verified minimal repair, that is, the smallest set of changed lines that, applied to the failing program, makes it pass every test. A deterministic classifier assigns this repair to one of twelve program-level categories, from loop boundaries to output format. The labels are therefore grounded in executed evidence rather than in judgement, and every one of them can be re-derived. Because real repairs are imbalanced and often mix several changes, we add a second benchmark of controlled single-edit mutants of accepted programs whose category is known by construction. With these benchmarks we answer four research questions:

RQ1 How much mechanism information does the test-outcome OAV lose?

RQ2 Do label-free descriptions of how the output deviates from the oracle

C-Pack-IPAs lab02-ex09: print a duration given in seconds as hh:mm:ss

Input 60, expected output 00:01:00\n. 78 of the 89 failing submissions fail all four tests.

Mechanism of the minimal repair	Verified minimal repair – failing + repaired	Tests 1–4	Printed (input 60)	Deviation OAV (aggregated)
(a) Output text	– <code>printf("%02d:%02d:%02d", ...);</code> + <code>printf("%02d:%02d:%02d\n", ...);</code>	✗✗✗✗	00:01:00	newline_end = missing whitespace_only = yes
(b) Output format	– <code>printf("%d:%d:%d\n", h,m,s);</code> + <code>printf("%02d:%02d:%02d\n", h,m,s);</code>	✗✗✗✗	0:1:0\n	numbers = format_only
(c) Computation	– <code>horas = n / 60 * 60;</code> + <code>horas = n / (60 * 60);</code> – <code>n = n % 60 * 60;</code> + <code>n = n % (60 * 60);</code>	✗✗✗✗	60:00:00\n	numbers = mixed
		identical	all different	

Figure 1: Three failing submissions to the same assignment (C-Pack-IPAs, anonymised) with an identical test signature. Their verified minimal repairs belong to three mechanism categories, and the aggregated deviation descriptors of Section 4.1 differ for each of them. Long `printf` arguments are elided; the printed output is shown for the first test, with `\n` for a newline.

recover mechanisms better at the same clustering budget?

RQ3 Do IF–THEN rules induced with the Inductive Learning Algorithm (ILA) and its noise-tolerant variant ILA-2 (Tolun and Abu-Soud, 1998; Tolun et al., 1999) predict the mechanism of new students, and do they carry over to new assignments?

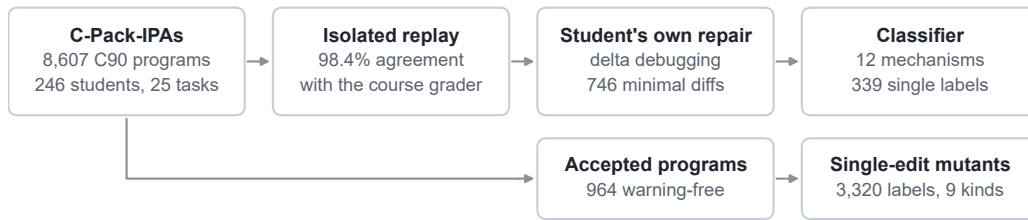
RQ4 Do frozen neural code embeddings cluster failing programs by mechanism or by the program they derive from?

Each question is tied to registered hypotheses with falsification criteria (Section 4.3); hypotheses, representations, weights and statistical rules were fixed before the evaluation, and every departure is recorded as a dated amendment. Figure 2 gives an overview of the study.

Our contributions are:

1. a reproducible, isolated replay of C-Pack-IPAs with an oracle audit against the historical grader (98.42% agreement over 28,081 test verdicts) and a second independent replay (99.68% agreement);
2. a repair-grounded benchmark built from 750 student repair events (746 verified pairs, 339 single-mechanism labels) and a controlled benchmark of 3,320 single-edit mutants, sharing one taxonomy and one classifier;

A Evaluation labels: built from executed evidence, never used as features



B Representations of a failing program: available before any repair



Instructor tool (AAI Lab, Fig. 8): groups with shared evidence, one ILA-2 rule per group, and a mechanism hypothesis only when a rule covers at least half of the group.

Figure 2: Overview of the study. (A) Evaluation labels come from executed evidence: the student’s own repair reduced by delta debugging, and single-edit mutants of accepted programs; they are never used as features. (B) Representations are computed from the failing program and its own runs only. The four research questions compare these representations against the labels, and the instructor tool of Section 6.1 applies the resulting design.

3. execution-deviation OAV, a label-free and problem-agnostic representation of output deviations, and an evaluation protocol with an identifiability ceiling and a granularity-matched null;
4. pre-registered evidence that test signatures mix mechanisms (at most 73.9% of real and 50.7% of injected items are identifiable from them), that output-deviation evidence recovers mechanisms better on controlled data but not reliably on real repairs, that rules over it transfer to unseen assignments, and that a frozen code embedding clusters programs by origin;
5. an instructor tool that applies these findings: it groups a class’s failing programs with the deviation OAV, attaches shared evidence and one induced rule to every group, and states a mechanism hypothesis only when a rule covers most of the group.

We release the replay runner, benchmarks, protocol, analysis code and tool. Throughout, a label describes what changed in the program; it is not a claim about what the student believed. Section 2 reviews related work, Section 3 builds the benchmarks, Section 4 defines representations, analyses and hypotheses, Section 5 answers the research questions, Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related work

Misconceptions and logic errors of novice programmers. Reviews of introductory programming describe misconceptions about variables, control flow, parameter passing and the notional machine that executes a program (Sorva, 2013; Qian and Lehman, 2017). Error studies distinguish slips from errors that follow from a wrong model of the language or task (Altadmri and Brown, 2015; Hermans, 2021). Ettles et al. (2018) classified logic errors of CS1 students into algorithmic errors, misinterpretations of the task and misconceptions, and found misconceptions to be the most frequent and the hardest to resolve. Misconception-driven feedback can help students when the misconception is known in advance (Gusukuma et al., 2018), and interactive instruments that ask students to predict program behaviour can elicit misconceptions directly (Lu and Krishnamurthi, 2024). These studies motivate automatic grouping of errors, but they also show that a program alone rarely reveals which belief produced it; we therefore evaluate program-level mechanisms and keep cognitive claims out of scope.

Clustering student programs. OverCode clusters correct solutions by normalised code (Glassman et al., 2015), and Clara clusters correct solutions by dynamic equivalence to repair incorrect ones (Gulwani et al., 2018). For

incorrect submissions, Kaleeswaran et al. (2016) cluster by repair strategy so that an instructor fixes one representative per cluster; MistakeBrowser clusters incorrect submissions by the code transformations that students applied when they fixed their own programs (Head et al., 2017); TraceDiff contrasts the traces of a buggy program and its closest fix (Suzuki et al., 2017). Program embeddings learned from code and execution have been used to propagate feedback (Piech et al., 2015), and Shi et al. (2021) clustered embeddings of incorrect block-based programs to help experts discover misconceptions. For C, InvAASTCluster combines dynamic invariants with abstracted syntax trees (Orvalho et al., 2025), and MENTOR couples clustering, formula-based fault localisation and LLM repair (Orvalho et al., 2026). These systems use repairs or correct solutions as part of the method. We use the student’s own later repair only to construct evaluation labels, and ask how well representations that are available *before* any repair recover them.

Mining misconceptions with learned models. Subtree-attention models localise logical errors from correctness labels alone (Hoq et al., 2025), pattern-based knowledge components are extracted from student code with representation learning and clustering (Hoq et al., 2026), and LLMs discover misconceptions from sets of student code samples in McMining, whose benchmark injects known misconceptions into Python code (Al-Hossami and Bunescu, 2026). Recent classroom tools cluster LLM-identified errors for instructors (Liding et al., 2026), ground tutoring in known misconceptions (Niousha et al., 2026), or refine automatically detected signals with expert review (Aruleba et al., 2026). PyMETA shows that diagnosis accuracy depends strongly on how the gold label is defined: prompted models that label the first execution error well agree far less with expert repair paths (Li et al., 2026). Our study shares this concern and makes the label definition explicit, verifiable and reproducible.

Benchmarks, mutants and repairs. C-Pack-IPAs provides C90 submissions of 25 assignments with anonymised student identifiers, tests and historical grading logs (Orvalho et al., 2024); ITSP is an earlier benchmark of buggy and corrected C programs (Yi et al., 2017). Mutation analysis injects controlled faults; mutant detection correlates with real-fault detection in software testing (Just et al., 2014), and conceptual mutants built from students’ wrong examples expose specification misconceptions (Prasad et al., 2024). Delta debugging isolates a minimal failure-inducing difference between a failing and a passing configuration (Zeller and Hildebrandt, 2002). We combine these ideas: we replay the corpus, reduce each student’s own repair to a verified minimal change, and pair these labels with controlled single-mechanism mutants.

Rule induction. The Inductive Learning Algorithm (ILA) extracts production rules class by class from attribute–value examples (Tolun and Abu-Soud, 1998); ILA-2 adds a penalty factor that tolerates noisy data (Tolun et al., 1999). ILA is sometimes listed alongside clustering methods, but it is supervised rule induction, and we use it as such: to state IF–THEN rules that map execution evidence to mechanisms and to measure how well such rules hold on students not used to induce them.

Positioning. Most systems above cluster to save instructor effort or to propagate feedback, and they are evaluated by the effort saved, by the correctness of generated repairs, or by expert inspection of clusters. To our knowledge, none measures how well a representation that is available before any repair recovers an independently verified error mechanism, and none reports how much of that mechanism the test-case signature can express at all. These are the two measurements this paper contributes, together with a benchmark that makes them repeatable.

3. Data and benchmark construction

3.1. Corpus and isolated replay

We use the canonical submissions of C-Pack-IPAs release 26 (Orvalho et al., 2024): 8,607 C90 programs by 246 anonymised students for 25 assignments of three laboratory classes, with 106 input/output tests. The historical logs record a verdict per test but often no output for passing tests, and 9,425 test cells were never executed or not logged. To obtain complete execution evidence we replayed every non-empty program in a research-only container: no network, read-only root file system, all Linux capabilities dropped except those needed to switch to an unprivileged per-run user, and resource limits of 3 s CPU, 5 s wall time and 512 MiB of address space per test. Programs were compiled with `gcc 12.2.0` and the course flags (`-ansi -pedantic -Wall -Wextra`); because the course also used `-Werror`, we recorded whether compilation was warning-free instead of rejecting the program. The container returns raw output bytes; oracles stay on the host, which decides every verdict.

The replay agreed with 98.42% of the 28,081 historical test verdicts under the exact-match, zero-exit policy (selected on the training partition among four comparison policies; Table 1), reproduced the logged output of 96.1% of historically wrong answers byte for byte, and matched the historical compilation outcome under `-Werror` semantics for 99.76% of programs. A second, independent replay from a clean commit agreed on 99.68% of test cells; the 56 programs whose outcomes differed between the two replays (likely undefined behaviour) were excluded from all items (amendment A5).

Table 1: Agreement (%) of the isolated replay with the historical per-test verdicts under four output comparison policies. The policy was chosen on the training partition before any mechanism label existed.

Comparison policy	Train	Validation	Test
Exact output, exit status 0 (selected)	98.23	99.04	98.52
Exact output, any exit status	98.16	98.70	98.36
Ignore trailing whitespace	89.03	89.75	91.13
Whitespace-separated tokens	83.88	86.22	86.47

3.2. Split

Submissions were split before any label existed into training, validation and test partitions (70/15/15% of connected components) such that all submissions of a student, across years and assignments, and all byte-identical sources fall into one partition. Every derived item inherits the partition of the submission it comes from.

3.3. Repair-grounded labels

For every failing submission (at least one test not passed) we looked for the same student’s next submission to the same assignment in the same year that passed every test. For each such repair event we kept the latest failing attempt before it (amendment A2), giving 750 failing–repaired pairs. Algorithm 1 summarises how one pair becomes a label. The line difference between the two programs, ignoring indentation, is split into hunks. Hunk-level delta debugging (Zeller and Hildebrandt, 2002) then searches for a 1-minimal subset of hunks whose application to the failing program makes it pass every test, executing every candidate in the isolated runner (budget 64 probes per pair). Both endpoints are re-verified in the same runner before the search.

The verified minimal change is classified into one of twelve program-level categories (Table 2) by a deterministic classifier that works on each verified line hunk: whole statements that were inserted or deleted are recognised after sliding the token run to statement boundaries, identical lines that were deleted and re-inserted elsewhere are placement changes, and every other changed token is attributed to its innermost role-defining syntactic context (loop header, branch condition, initialiser, output format, and so on). A pair receives a single-mechanism label only when its minimal repair falls into exactly one category other than OTHER; repairs spanning several categories are kept as MULTI and excluded from the primary gold. The repairing submission, the patch and the label are never used as features. Figure 3A traces every exclusion.

Algorithm 1 Repair-grounded labelling of one repair event

Require: failing program f ; the same student’s next program p that passes every test of suite T

- 1: run f and p on T in the isolated runner; stop unless f fails a test and p passes all ▷ endpoints
 - 2: $H \leftarrow$ line hunks of $\text{diff}(f, p)$, indentation ignored
 - 3: $H^* \leftarrow \text{ddmin}(H)$, where a subset $S \subseteq H$ succeeds iff f patched with S compiles and passes all of T
 - 4: $C \leftarrow \{\text{category}(h) : h \in H^*\}$ ▷ syntactic role of each changed token
 - 5: **if** $C = \{c\}$ and $c \neq \text{OTHER}$ **then**
 - 6: **return** label c
 - 7: **else**
 - 8: **return** MULTI or OTHER (not in the gold)
 - 9: **end if**
-

3.4. Controlled mechanism injection

Real repairs are ecologically valid but noisy, imbalanced and specific to what students happened to do. We therefore also built a controlled benchmark. From accepted, warning-free programs (at most 40 per assignment, one per student and year) we generated single-edit mutants with operators aligned with the same taxonomy: strictness of a loop or branch comparison, logical connective, loop start, constant initialiser, arithmetic operator, removed float cast or float literal, printf precision, dropped trailing new-line, letter case of output text, deleted statement, duplicated statement, and statements moved into a loop. One random site per category and source was mutated with a fixed seed. A mutant was kept only if it compiled without warnings and failed at least one test, that is, if it could have been a real failing submission of the course (Figure 3B). A unit test checks that the classifier of Section 3.3 assigns every operator’s inverse edit to the operator’s category, so both label sets share one definition.

4. Methods

4.1. Object–attribute–value representations

Every failing program (the object) is described by categorical attributes computed from the program and its own replayed runs only. Let T be the tests of an assignment, y_t the expected output of test t , and $\hat{y}_t(o)$ the output of program o . The test-outcome OAV of the classic formulation is

$$x^{\text{out}}(o) = (r_t(o))_{t \in T}, \quad r_t(o) \in \{\text{pass, fail, runtime error, timeout}\}. \quad (1)$$

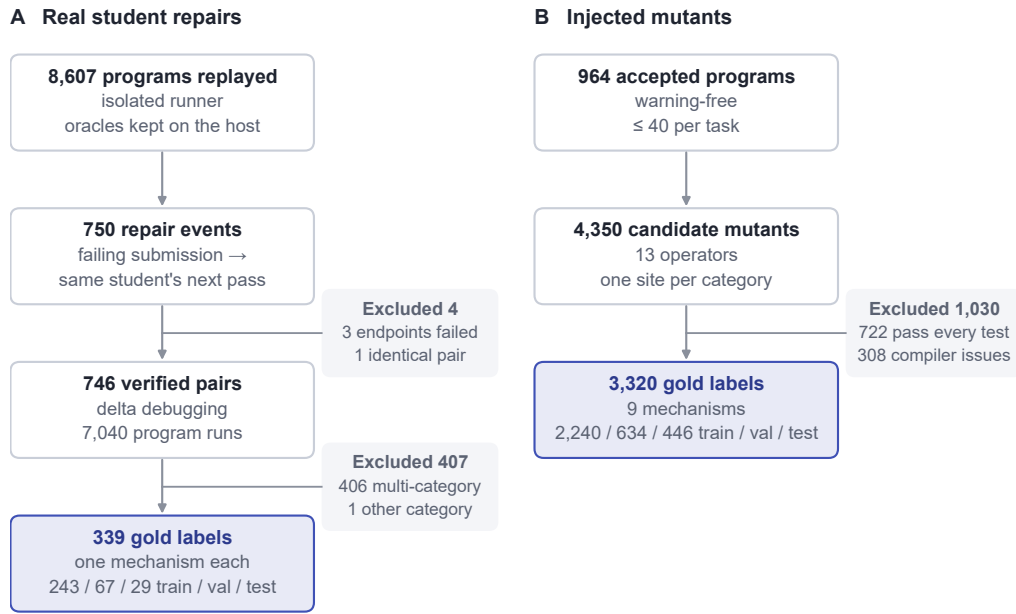


Figure 3: Construction of the two benchmarks. (A) Real student repairs: 743 of the 746 verified pairs reached a 1-minimal subset within the probe budget; the other three spanned several categories. (B) Injected mutants: candidates that passed every test (equivalent on the suite) or did not compile cleanly were discarded. Partition sizes are training / validation / test.

Table 2: Program-level mechanism taxonomy. A category describes what the verified minimal repair (or the injected edit) changes; the right column lists misconception hypotheses an instructor may check, not diagnoses.

Category	The minimal change edits...	Hypotheses to check
Loop boundary	condition, start or update of a loop	off-by-one, 0- vs 1-based ranges
Branch condition	condition or structure of if/switch/else	boundary of comparisons, <code>&&</code> vs <code> </code>
Initialisation	a declaration initialiser or constant assignment outside loops	accumulator or extremum initialisation
Computation	operators or operands of assignments, returns, printed expressions	formula, precedence
Numeric type	types, casts, float literals, math functions	integer division
Output format	printf conversion specifications	precision, width
Output text	literal output text or print-only statements	reading the output specification
Input	scanf/getchar calls	input format, end of file
Statement placement	a moved statement or brace	scope of blocks, reset inside loops
Missing statement	inserted whole statements	missing update or case
Extra statement	deleted whole statements	duplicated update, debug output
Control flow	break/continue/return	premature exit

The *deviation OAV* adds, for every test, $K = 9$ label-free descriptors $d_k(\hat{y}_t(o), y_t)$ of how the output deviates from the expected one—run status, line count, trailing newline, whitespace-only difference, relation between corresponding numbers (off by one, truncated, precision, sign, zero, scale, format only), relation between words (case only, typo, missing or extra words), position of the first differing line, prefix relations, and equality with another test’s expected output—together with their mode over the failing tests,

$$\bar{d}_k(o) = \text{mode}\{d_k(\hat{y}_t(o), y_t) : r_t(o) \neq \text{pass}\}, \quad (2)$$

and the fraction of failing tests. The aggregated block $\bar{d}(o)$ does not depend on the assignment, so it can be compared across assignments; Figure 1 shows

its values for three programs with one test signature. Vocabularies are fitted, without labels, on training-partition failing submissions of the same assignment; values unseen in training are mapped to *unknown*. We compare eight representations:

- **Outcomes:** x^{out} .
- **Structural:** presence of 16 C constructs (loop kinds, inclusive/strict comparisons, subscripts, pointers, ...).
- **Code cues:** the eighteen static cues of the evidence representation below, alone.
- **Outcomes+stdout** and **Combined+stdout:** the strongest representations of our earlier pipeline, adding per-test output relations (exact, whitespace, empty, another test’s oracle, different) and edit-distance bands, and in the combined case also the structural block.
- **Deviation** (proposed): x^{out} with the per-test descriptors d_k and the aggregated block $\bar{d}$.
- **Evidence** (proposed): deviation plus eighteen static code cues (strict and inclusive loop comparisons, non-zero loop start, constant initialisers, division without a float operand, casts, printf precisions, output inside loops, output words absent from every oracle, ...).
- **Embedding:** a frozen code embedding of the raw source (Qwen3-Embedding-0.6B, Zhang et al. 2025), L2-normalised.

Block masses were fixed in the registered protocol before any evaluation and were not tuned (outcomes 1; outcomes+stdout 0.6/0.4; combined+stdout 0.48/0.12/0.4; deviation 0.3 tests, 0.4 per-test deviation, 0.3 aggregate; evidence 0.2/0.3/0.2 and 0.3 code cues). Within a block of mass m_b the mass is shared equally by the $|A_b|$ attributes that vary in training, $w_a = m_b/(|A_b| \sum_{b'} m_{b'})$, and the distance between two programs is the weighted Hamming distance

$$\delta(o, o') = \sum_a w_a [x_a(o) \neq x_a(o')]. \quad (3)$$

4.2. Clustering and rules

Each assignment’s failing programs form a cohort that is clustered transductively. Agglomerative clustering (HAC) uses average linkage on δ . K-means runs on the one-hot encoding scaled by $\sqrt{w_a/2}$, whose squared Euclidean distance equals δ , with seeds 7, 42 and 91; embeddings use Euclidean

K-means and cosine distance. The primary analysis gives every method the number of gold categories present in the cohort (*oracle k*, capped by the number of distinct representation patterns), which isolates the representation from model selection; a secondary analysis selects $k \in [2, 10]$ by silhouette without labels. The cheapest baseline groups identical test signatures.

For IF-THEN rules we use ILA (Tolun and Abu-Soud, 1998) and ILA-2 (Tolun et al., 1999) over problem-agnostic attributes: the aggregated deviation descriptors and code cues (*evidence*), or only the aggregated outcome state and the fraction of failing tests (*outcome summary*). Both algorithms induce rules class by class from combinations of at most two attribute-value conditions that cover at least three training examples. ILA keeps the combination that covers the most not-yet-covered examples of the class and no example of another class; ILA-2 scores a combination as its true positives minus a penalty factor times its false positives, so a rule may cover a few examples of other classes when it covers many of its own, which tolerates label noise. Rules are induced from training-partition labels, applied in precision order, and examples matched by no rule are counted as abstentions and errors. The ILA-2 penalty factor was the only quantity tuned, on the validation partition among $\{1, 2, 5\}$ (it selected 1.0). We compare with a depth-4 CART tree on the same attributes and with the majority class, on seen assignments with unseen students, and on unseen assignments (laboratory 4) with unseen students.

4.3. Measures and registered hypotheses

The primary measure is the adjusted Rand index (ARI; Hubert and Arabie 1985) between a cohort’s clusters and its gold categories, macro-averaged over cohorts with at least six items and two categories; we also report adjusted mutual information (Vinh et al., 2010), pairwise F1 and purity. The *identifiability ceiling* of a representation is $\sum_b \max_y n_{b,y}/N$ over its buckets b of identical attribute vectors: no clustering or classifier that only reads the representation can exceed it. Because deviation OAV refines the outcome OAV, its ceiling cannot be lower; we therefore compare it with the expected ceiling of a random refinement that splits every outcome bucket into sub-buckets of the same sizes (200 permutations).

The protocol, registered before any mechanism evaluation, states five hypotheses with falsification criteria (seven dated amendments were recorded before the sealed test run, six of them before any evaluation result; Appendix Appendix A). They map to the research questions as follows:

- RQ1: H1a, identical test signatures mix mechanisms (the upper 95% bound of the mean outcome ceiling is below 0.95); H1b, the refinement

added by deviation OAV carries mechanism information beyond its granularity.

- RQ2: H2, deviation OAV yields higher ARI than outcome OAV; H3, evidence OAV yields higher ARI than the strongest earlier baseline, combined+stdout.
- RQ3: H4, ILA-2 rules over agnostic evidence reach a higher test macro-F1 than rules over the outcome summary on the real benchmark.
- RQ4: H5, on the injected benchmark, frozen code-embedding clusters align more with the source program than with the injected mechanism.

Differences are summarised by their mean over cohorts with a 95% bootstrap interval (10,000 resamples) and Wilcoxon signed-rank tests with Holm correction; rule comparisons use a bootstrap over students. A hypothesis is supported only when the interval of the difference excludes zero in the stated direction on every benchmark it names. Because real single-mechanism items are sparse in the sealed test partition (0 eligible cohorts), the registered amendment A3 evaluates clustering on the real benchmark over full per-assignment cohorts (clustering fits nothing on labels); the injected benchmark is evaluated on test-partition cohorts, with full cohorts reported in the released results. Rule induction always trains on the training partition and is scored on the test partition.

5. Results

5.1. Benchmarks

Of the 750 repair events, 746 passed the endpoint checks and 743 reached a 1-minimal repair within the probe budget; 72% of minimal repairs changed one or two line hunks (Figure 3A). 54.4% of the minimal repairs spanned more than one category and were excluded from the gold; the remaining 339 single-mechanism items are dominated by output text (the course graded output byte for byte), followed by computation, missing statements, branch conditions, loop boundaries, output format and control flow (7 of the 8 audited control-flow repairs added the `return 0` that C90 needs for a zero exit status). The controlled benchmark contains 3,320 mutants over nine categories; 16.6% of candidate mutants passed every test and were discarded as equivalent on the suite (Figure 3B). Table 3 summarises the registered hypotheses; the following subsections answer the research questions in turn.

Table 3: Registered hypotheses: mean difference over cohorts, 95% bootstrap interval and verdict (H1a reports the mean outcome ceiling itself). H2 and H3 are reported for K-means and HAC.

Hypothesis	Real repairs			Injected mutants		
	Mean	95% CI	Verdict	Mean	95% CI	Verdict
H1a	0.739	[0.696, 0.783]	supported	0.507	[0.429, 0.588]	supported
H1b	0.080	[0.052, 0.112]	supported	0.172	[0.121, 0.224]	supported
H2 kmeans	0.054	[−0.081, 0.182]	not supported	0.235	[0.143, 0.331]	supported
H2 hac	0.086	[−0.025, 0.200]	not supported	0.214	[0.117, 0.314]	supported
H3 kmeans	−0.006	[−0.097, 0.117]	not supported	0.002	[−0.055, 0.067]	not supported
H3 hac	−0.010	[−0.111, 0.119]	not supported	−0.010	[−0.060, 0.041]	not supported
H5	–	–	–	0.834	[0.695, 0.958]	supported

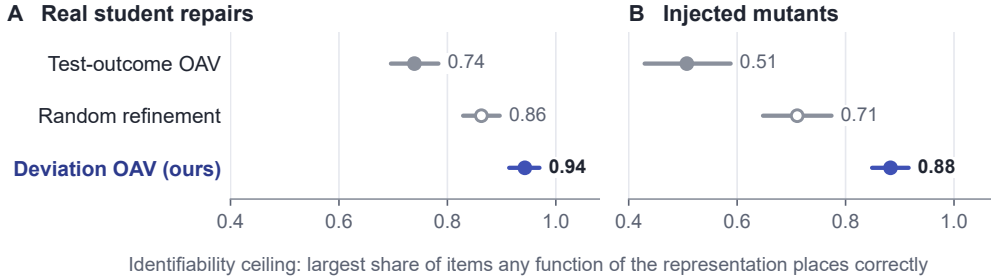


Figure 4: RQ1. Identifiability ceiling of the test-outcome OAV, of a random refinement that splits every outcome bucket into sub-buckets of the same sizes as deviation OAV (open circles), and of deviation OAV. Points are means over cohorts; bars are 95% bootstrap intervals.

5.2. RQ1: test signatures mix mechanisms

Identical test signatures often hide different mechanisms (Figure 4). The best any function of the test-outcome OAV can do is to assign 73.9% of real items (mean ceiling 0.74, 95% CI [0.70, 0.78]) and 50.7% of injected items ([0.43, 0.59]) to their mechanism, so H1a is supported on both benchmarks. Put differently, 35.5% of the 2,087 real pairs that share a test signature within an assignment have different mechanisms, and 81.6% of the 2,901 injected pairs do. The refinement added by deviation OAV raises the ceiling by 0.080 [0.052, 0.112] on real and 0.172 [0.121, 0.224] on injected cohorts beyond a random refinement of the same granularity (H1b supported; Holm-adjusted $p = 0.0018$ and $p < 0.0001$). The extra distinctions therefore carry mechanism information rather than mere resolution.

5.3. RQ2: clustering by mechanism

Table 4 and Figure 5 report mechanism recovery at the same clustering budget. On the controlled benchmark, clusters of the test-outcome OAV

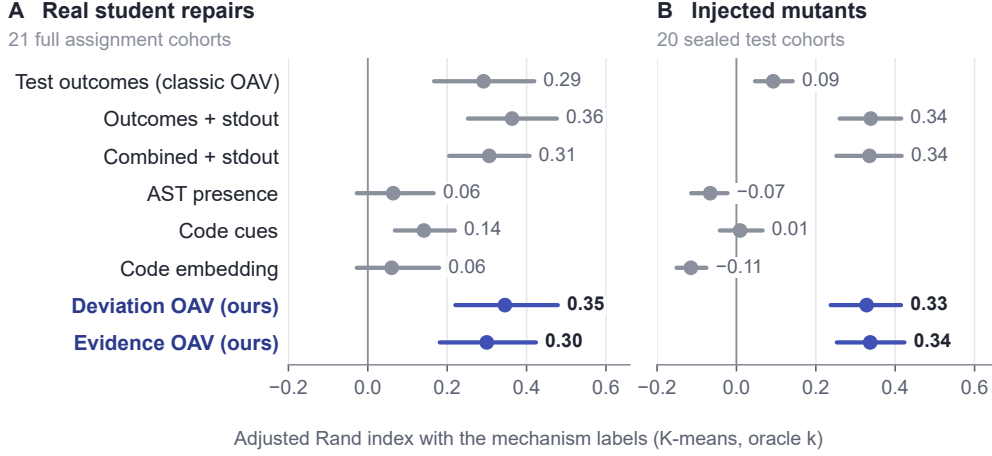


Figure 5: RQ2. Adjusted Rand index between K-means clusters (oracle k) and mechanism labels. Indigo: proposed representations; grey: baselines. Points are means over cohorts; bars are 95% bootstrap intervals over cohorts. Real repairs are evaluated on full assignment cohorts (amendment A3), injected mutants on sealed test cohorts.

barely agree with mechanisms (ARI 0.09, exact signatures 0.10), while deviation OAV reaches 0.33: a difference of 0.235 [0.143, 0.331] with K-means and 0.214 [0.117, 0.314] with HAC, larger in 16/20 cohorts. On real repairs the test-outcome OAV already reaches 0.29 because the dominant output-text category produces distinctive signatures; deviation OAV reaches 0.35 with K-means and 0.38 with HAC, but the differences (0.054, CI [-0.081, 0.182]; 0.086, CI [-0.025, 0.200]) are not reliable. H2 is therefore supported on the controlled benchmark only and, by the registered rule, not supported overall.

H3 is not supported: evidence OAV and the earlier combined+stdout baseline are indistinguishable on both benchmarks (differences -0.006 and 0.002). Simple output relations already capture most of what distance-based clustering can exploit; outcomes+stdout reaches 0.36 and 0.34. Selecting k by silhouette instead of the oracle lowers all scores slightly and preserves the ordering (Table 4).

Which mechanisms benefit (exploratory). A cohort-level ARI weighs large categories most, so we also measure, per category, the share of items whose K-means cluster has that category as its majority label (Figure 6). On real repairs, test signatures recover output text (0.98), loop boundaries (0.88) and control flow (0.83), but not output format (0.00), branch conditions (0.24) or missing statements (0.37); with deviation OAV these rise to 0.69, 0.59 and 0.56. Averaged over categories, recovery rises from 0.54 to 0.71 on real and from 0.50 to 0.72 on injected cohorts. The gains of output

Table 4: Mechanism recovery on the two benchmarks: mean ARI over cohorts (oracle k unless stated) and mean identifiability ceiling. Real repairs: full per-assignment cohorts (amendment A3); injected mutants: sealed test-partition cohorts.

Representation	K-means	HAC	K-means, silh. k	Ceiling
<i>Real repairs</i>				
Test outcomes (classic OAV)	0.292	0.292	0.239	0.739
Outcomes + stdout	0.363	0.367	0.313	0.907
Combined + stdout	0.306	0.334	0.314	0.980
AST presence	0.064	0.029	0.080	0.772
Code cues	0.141	0.157	0.105	0.931
Code embedding	0.060	0.077	0.063	—
Deviation OAV (ours)	0.345	0.378	0.295	0.943
Evidence OAV (ours)	0.300	0.324	0.279	0.986
Exact test signature	0.281	—	—	—
<i>Injected mutants</i>				
Test outcomes (classic OAV)	0.093	0.084	0.088	0.507
Outcomes + stdout	0.338	0.299	0.332	0.839
Combined + stdout	0.335	0.293	0.309	0.914
AST presence	−0.066	−0.068	−0.052	0.353
Code cues	0.009	−0.009	−0.020	0.654
Code embedding	−0.115	−0.104	−0.114	—
Deviation OAV (ours)	0.328	0.297	0.328	0.883
Evidence OAV (ours)	0.337	0.283	0.227	0.964
Exact test signature	0.096	—	—	—

evidence concentrate in the minority mechanisms, which is why the real-benchmark ARI difference is small. This analysis was added after the sealed run (amendment A7) and is not a registered hypothesis.

5.4. RQ3: IF-THEN rules

ILA-2 rules over problem-agnostic evidence attributes predicted the mechanism of real test-partition items with macro-F1 0.26 (coverage 0.69, accuracy on answered items 0.70), against 0.13 for rules over the outcome summary and 0.11 for the majority class (Figure 7 and Table 5). The difference, 0.094 [−0.011, 0.171], rests on the 29 single-mechanism items of the sealed test students and includes zero, so H4 is not supported.

On the injected benchmark, where every category is represented, evidence rules reached macro-F1 0.65 on seen and 0.35 on unseen assignments, whereas outcome-summary rules reached 0.03 and 0.04: rules over the outcome state alone hardly transfer, while rules over output deviations do. The attribute

A Real student repairs

	Test outcomes	Combined + stdout	Deviation OAV	Evidence OAV
Output text (177)	0.98	0.92	0.95	0.94
Computation (28)	0.54	0.64	0.57	0.61
Missing statement (27)	0.37	0.67	0.56	0.63
Branch condition (17)	0.24	0.65	0.59	0.59
Loop boundary (16)	0.88	0.94	0.81	0.81
Output format (13)	0.00	0.69	0.69	0.69
Control flow (12)	0.83	0.83	0.83	0.92
Extra statement (8)	0.12	0.38	0.50	0.50
Initialisation (7)	0.57	1.00	1.00	1.00
Input (6)	0.50	0.50	0.50	0.50
Statement placement (6)	0.50	0.33	0.50	0.33
Numeric type (3)	1.00	1.00	1.00	1.00

B Injected mutants

	Test outcomes	Combined + stdout	Deviation OAV	Evidence OAV
Missing statement (72)	0.49	0.69	0.60	0.64
Extra statement (65)	0.46	0.68	0.82	0.82
Output text (60)	0.62	0.90	0.97	0.95
Initialisation (53)	0.40	0.60	0.53	0.58
Loop boundary (50)	0.56	0.68	0.62	0.62
Statement placement (48)	0.08	0.67	0.79	0.58
Computation (43)	0.51	0.70	0.67	0.77
Branch condition (29)	0.34	0.45	0.45	0.48
Output format (9)	1.00	1.00	1.00	0.89

Share of items whose K-means cluster has their mechanism as majority label

Figure 6: RQ2, exploratory. Share of items of each mechanism whose K-means cluster (oracle k , seed 42) has that mechanism as its majority label; numbers of items in parentheses. Darker cells recover more.

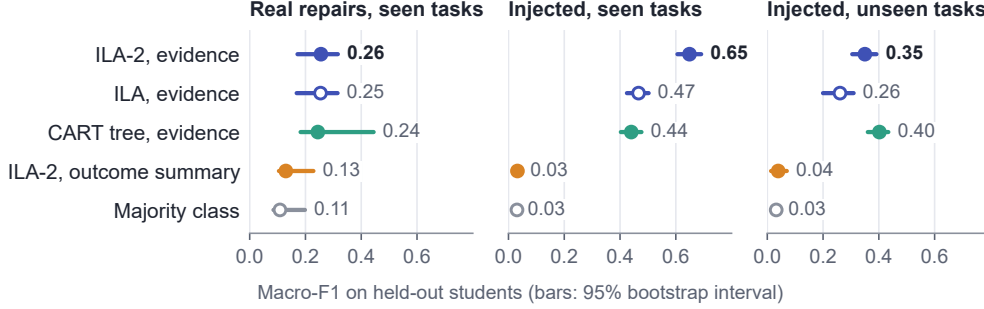


Figure 7: RQ3. Macro-F1 of rule learners on held-out students, with abstentions counted as errors; bars are 95% bootstrap intervals over students. *Evidence*: aggregated deviation descriptors and code cues; *outcome summary*: aggregated outcome state and fraction of failing tests. Unseen tasks are the assignments of laboratory 4, absent from training.

set matters more than the learner. A CART tree on the same evidence attributes reached 0.44 and 0.40, below ILA-2 on seen assignments and slightly above it on unseen ones. The noise-tolerant penalty of ILA-2 traded a little precision for much coverage: the original ILA answered 0.34 of injected test items with accuracy 0.94, ILA-2 answered 0.81 with accuracy 0.74. Table 6 lists the highest-precision rules learned from real repairs; they read like the production rules of the classic formulation, but their conclusions are program-level mechanisms and their precision was measured on students not used to induce them.

5.5. RQ4: frozen code embeddings cluster programs, not mechanisms

Qwen3-Embedding clusters of injected mutants agree more with the accepted program the mutant came from than with the injected mechanism, by 0.834 ARI (95% CI [0.695, 0.958]; H5 supported). Their mechanism ARI is -0.11 on injected and 0.06 on real cohorts (Figure 5). Single-token changes that decide a program’s behaviour barely move a general-purpose embedding, whereas authorship and solution style dominate it.

6. Discussion

What test signatures can and cannot tell an instructor. The classic pipeline—test-case OAV, K-means or Hamming clustering, induced IF-THEN rules—recovers the most frequent failure of this course well: output-text errors produce distinctive signatures, and on real repairs the signature clusters reach a moderate ARI of 0.29. But the representation cannot separate mechanisms that fail the same tests. On controlled mutants of the same programs, where every mechanism is present, its clusters barely agree with mechanisms (ARI

Table 5: IF–THEN rules predicting the mechanism of test-partition students from problem-agnostic attributes, real repairs. Abstentions count as errors in macro-F1; selective accuracy is measured on answered items.

Track	Model (attributes)	Macro-F1	Coverage	Selective acc.
Seen assignments	ILA (evidence)	0.254	0.586	0.765
Seen assignments	ILA-2 (evidence)	0.256	0.690	0.700
Seen assignments	CART (evidence)	0.244	1.000	0.621
Seen assignments	ILA (outcome summary)	0.000	0.000	0.000
Seen assignments	ILA-2 (outcome summary)	0.130	0.828	0.583
Seen assignments	CART (outcome summary)	0.251	1.000	0.586
Seen assignments	Majority class	0.109	1.000	0.483
Unseen assignments	ILA (evidence)	0.354	1.000	0.500
Unseen assignments	ILA-2 (evidence)	0.354	1.000	0.500
Unseen assignments	CART (evidence)	0.381	1.000	0.500
Unseen assignments	ILA (outcome summary)	0.000	0.000	0.000
Unseen assignments	ILA-2 (outcome summary)	0.214	0.750	0.500
Unseen assignments	CART (outcome summary)	0.214	1.000	0.375
Unseen assignments	Majority class	0.136	1.000	0.375

0.09), and even a perfect classifier reading only the signature could assign at most 50.7% of items correctly. A dashboard built on signatures therefore reports the common, often specification-level failures accurately and merges the less frequent conceptual ones—wrong branch conditions, missing steps, output format—that are arguably the reason to cluster in the first place.

Output evidence is the cheapest improvement, not new models. Everything that improved mechanism recovery was already in the autograder’s output: relations between actual and expected output and, in our deviation OAV, the kind of numeric and textual deviation. On controlled data this more than tripled the ARI at the same clustering budget, and on real repairs it lifted per-category recovery of the minority mechanisms. Our richer descriptors did not beat simple output relations for clustering (H3), so we do not claim a better clustering method. Their value lies elsewhere: they are problem-agnostic, so rules learned on some assignments apply to others. Rules over aggregated deviation descriptors kept a macro-F1 of 0.35 on unseen assignments of the controlled benchmark, where rules over outcome summaries were near zero; per-test signatures cannot even be compared across assignments.

Rules as checked hypotheses. ILA and ILA-2 produce the production rules that the classic formulation asks for, and ILA-2’s penalty trades a small loss of precision for much larger coverage. A shallow CART tree over the same attributes is about as accurate, so the case for ILA-2 is not accuracy but form: an ordered list of short IF–THEN rules, each with a support count,

Table 6: The most precise ILA-2 rules induced from real training repairs (support: correct/matched training items).

IF	THEN nism	Support
the program has no output call	missing statement	10/10
the program has no input call	missing statement	8/8
printed numbers equal the oracle’s but are written differently	output format	8/8
the trailing newline is as expected and fewer numbers than the oracle are printed	output text	7/7
the failing tests time out	loop boundary	4/4
the output is a truncated prefix of the oracle	output text	78/79
the output differs from the oracle only in whitespace	output text	97/107
the output contains words the oracle lacks	output text	20/24

that an instructor can read and falsify, and that abstains instead of guessing when no rule matches. On real repairs the advantage of evidence rules over outcome-summary rules was not statistically reliable with the 29 sealed test items, so these rules should be read as starting points for an instructor’s judgement, not as diagnoses.

Embeddings find authors before mechanisms. A strong general-purpose code embedding grouped mutants by the program they came from: one-token changes that decide behaviour barely move the vector, while identifiers, layout and solution strategy dominate it. This does not contradict earlier successes with embeddings trained on the task (Shi et al., 2021; Piech et al., 2015); it shows that a frozen embedding needs execution evidence or mechanism-aware training before its clusters can be read as error groups.

6.1. Implications for instructors and tool builders

We turned these findings into a local instructor tool, AAI Lab (Figure 8), whose design follows from the evidence rather than from the classic formulation. For every exercise, the latest failing submission of each learner is represented with the registered deviation OAV and clustered with K-means, with k chosen by silhouette unless the instructor sets it. Each group shows (i) its test signature and, for every test, the failure rate of the group next to that of the other groups, (ii) the output deviations shared by at least 60% of its members, and (iii) one IF–THEN rule induced by ILA-2 on the class itself, reported with its precision and coverage. A mechanism hypothesis is stated

only when one rule covers at least half of the group: either a hand-written rule for that exercise or for C programs in general, or one of the 14 ILA-2 rules learned from real repairs and frozen before deployment. Otherwise the group is labelled as having no hypothesis, and when its members match different rules the tool says so explicitly, as for the programs of Figure 1. Every hypothesis comes with one question to ask students and one re-teaching activity, and the instructor confirms or rejects the group; programs run only in an isolated container, and the tool serves the local machine only.

Three lessons from the study shaped this design. First, a group is described by what its members share, never by the signature alone, because RQ1 showed that signatures merge mechanisms. Second, the tool abstains and flags mixed groups rather than naming every cluster, because RQ3 showed that even the best rules leave many items unexplained on real data. Third, mechanism hypotheses are phrased as questions to check with students, because a repair-grounded label describes a program change, not a belief. We have not yet evaluated the tool with instructors; whether its groups and hypotheses change teaching decisions is an open question for a classroom study.

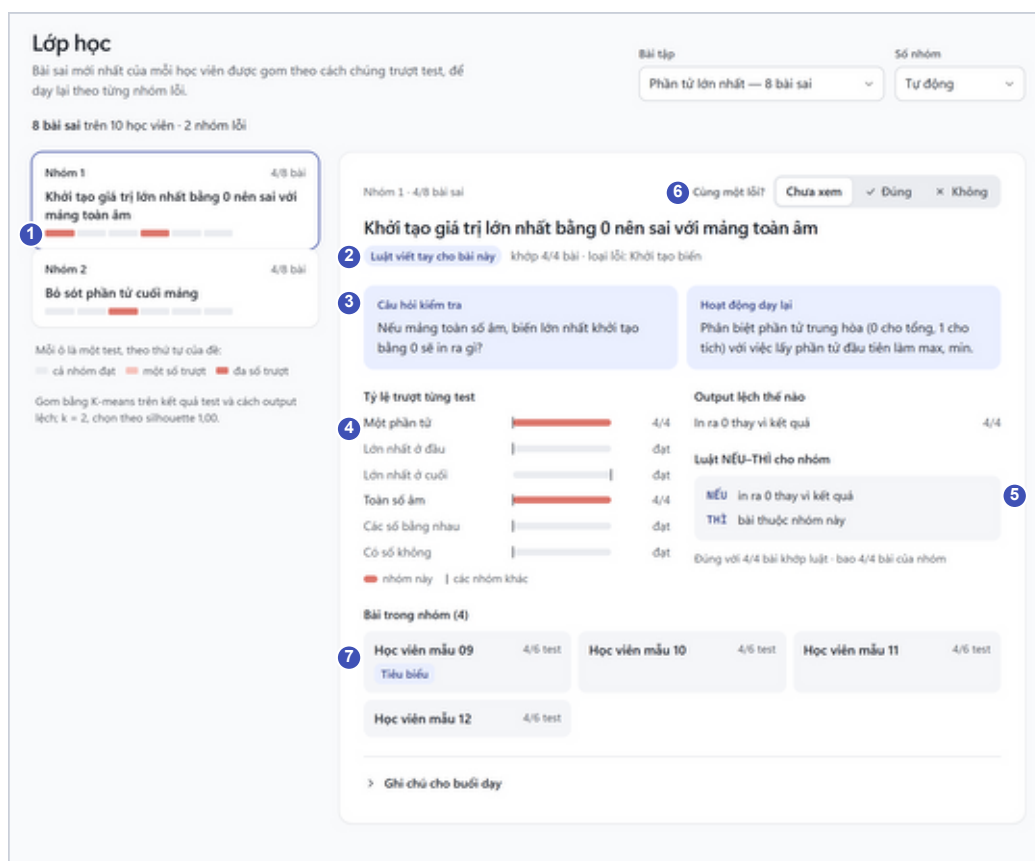
6.2. *Repair-grounded evaluation*

Verified minimal repairs turned 750 naturally occurring repair events into auditable labels at the cost of 7,040 program executions and no annotation hours. Two findings matter beyond this study. First, 54.4% of minimal repairs changed more than one kind of code, so partitioning failed submissions into single-mechanism clusters is itself a simplification; overlapping or multi-label grouping deserves more attention. Second, a controlled benchmark built with the same taxonomy exposes limits that the imbalanced real distribution hides, and the two benchmarks disagree exactly where the dominant real category is easy.

6.3. *Limitations and threats to validity*

Labels describe programs, not minds. A repair-grounded label records what the student changed so that the program passes; the same change can follow from a misconception, a slip or a misread specification, and a student may fix a symptom without revising a belief. We therefore evaluate program-level mechanisms and phrase teaching output as hypotheses to check. Claims about beliefs require think-aloud, explanation or prediction tasks with students (Lu and Krishnamurthi, 2024).

Label construction. Delta debugging works on line hunks, so two changes on one line are inseparable and a minimal repair can include a formatting edit



- 1 Test signature of each group (one cell per test)
- 2 Hypothesis: rule source and members covered
- 3 A question to ask and a re-teaching activity
- 4 Per-test failures: group bar vs. other groups
- 5 ILA-2 rule for the group, precision and coverage
- 6 The instructor confirms or rejects the group
- 7 Members, representative (medoid) first

Figure 8: The class view of AAI Lab for one exercise of a demonstration class written by the authors (interface in Vietnamese). The selected group of four submissions fails the tests with a single element and with only negative numbers; the hand-written exercise rule “largest value initialised to 0” covers all four, and the rule induced on the class reads “IF the output is 0 instead of the result”.

adjacent to a logic fix. The classifier is rule-based and was revised once after an audit of training and validation items (amendment A6); both audits were performed by an AI assistant and are not independent human annotation. Excluding multi-category repairs (MULTI, 54.4% of repair events) keeps the gold precise but biases it towards simple, often output-level, mechanisms. We treat attempts `sub_NNN` as chronological within a student, assignment and year.

Injected mutants. Mutants are cleaner than real errors, their category distribution is chosen by us, and code-based features inspect the edited syntax (amendment A4). Mutation-based evaluation is informative about what a representation *can* separate (Just et al., 2014), not about how often mechanisms occur in class.

Scope. One course, one language standard (C90), one grader policy (exact output match) and one institution. Deviation descriptors are problem-agnostic by design, but their value depends on output-based grading; courses with property-based or unit tests may see different patterns. The sealed test partition of the real benchmark was too small for cohort-level clustering statistics, which is why amendment A3 evaluates full cohorts; rule induction always used held-out students.

Statistics. Cohorts differ in size and label balance; ARI is sensitive to dominant categories. We report cohort-level bootstrap intervals, Wilcoxon tests with Holm correction and silhouette-based k as a secondary analysis, but the number of cohorts (tens) limits power, and items within a cohort are not independent across representations of the same program.

The tool. AAI Lab has been exercised on a demonstration class of programs written by the authors and on the C-Pack cohorts, not with instructors or students. Its hand-written exercise rules cover six exercises only, and its thresholds (60% for shared evidence, 50% for a hypothesis) are design choices that were not tuned or validated.

7. Conclusion

Clustering failed programs by the tests they fail is an appealing way to give instructors macro-feedback, and its IF-THEN rules read like teaching advice. Evaluated against verified minimal repairs of the students’ own programs and against controlled single-mechanism mutants, the test-case OAV turned out to mix mechanisms systematically: it identifies the dominant output-level failures and merges much of the rest (RQ1). Label-free evidence

about how the output deviates from the oracle recovers more, most clearly where mechanisms are balanced (RQ2), and it supports rules that carry across assignments (RQ3); general-purpose code embeddings, in contrast, cluster programs by origin (RQ4). Because every label in our benchmarks is derived from executed evidence and can be re-derived, the benchmarks can be reused to evaluate new representations, LLM-based diagnosers or active tests, and the instructor tool shows how the findings translate into feedback that abstains when the evidence is weak.

Three steps would strengthen these conclusions: an independent audit of the labels by two human raters, replication on other courses, languages and grading policies, and a classroom study of whether mechanism hypotheses change what instructors re-teach. Adaptive tests that break the remaining signature collisions before a hypothesis reaches the instructor are a natural next method. Claims about students’ beliefs need studies with students; the tools released here are meant to make such studies cheaper to design.

Data and code availability

The replay runner, benchmark construction scripts, registered protocol with its amendments, analysis code, figure scripts and the AAI Lab tool are available in the project repository (<https://github.com/linhlinhlin/AAI>). The benchmarks are derived from the public C-Pack-IPAs corpus (Orvalho et al., 2024); they are released as item identifiers, verified patches and labels, so that the programs themselves are obtained from the original release.

Ethics statement

The study uses only the public, anonymised C-Pack-IPAs corpus; no new data were collected from students or instructors, and no human participants took part. The demonstration class shown in Figure 8 consists of programs written by the authors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to write and review analysis code, to audit samples of the benchmark labels (as reported in Section 6.3), and to draft and edit the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Registered protocol and amendments

The protocol (`research/protocol-mechanism-v1.json`) was committed at 2026-09-30 12:06:00 UTC before any mechanism evaluation. Amendments A1–A6 were recorded before any evaluation result was viewed and A7 before the sealed test run; each is dated in the protocol file.

- A1** The original H1 compared the ceilings of the deviation and outcome representations. Because the deviation representation refines the outcome representation, that comparison could not fail; it was replaced by H1a (absolute outcome ceiling) and H1b (gain over a granularity-matched random refinement).
- A2** One item per repair event: the latest failing attempt before the repairing submission, since earlier attempts of the same student are near-duplicates.
- A3** If fewer than ten test-partition cohorts of the real benchmark were eligible, its clustering hypotheses would use full per-assignment cohorts, because clustering fits nothing on labels. The condition held (0 eligible test cohorts).
- A4** Code-based features inspect the syntax that injection operators edit, so their injected scores are an optimistic bound.
- A5** Programs whose outcomes differed between the two independent replays were excluded.
- A6** Classifier v1.1 after a label audit of training and validation items (below). A validation run started before this amendment was stopped before it wrote any result.
- A7** The ILA-2 penalty was frozen at 1.0, the value with the best validation macro-F1 on both benchmarks, and per-item assignments of the primary configuration were stored for exploratory per-category analyses.

Appendix B. Label audit

Two stratified samples of training and validation items were reviewed by the analysis assistant (an AI model; this is not an independent human annotation). The first sample (64 cases, up to five per assigned label) found all 53 single-category labels correct and 6 of 11 excluded cases to have a clear category under the taxonomy (initial values added to declarations, a type change by moving a declarator, a macro constant, and a wholly missing implementation). Classifier v1.1 handles these patterns; re-labelling the stored repairs changed 25 primary labels. A second, disjoint sample of 29 cases of the v1.1 labels agreed in 29 cases. Over both samples, the single-category labels agreed in 79/79 cases (Wilson 95% lower bound 0.954). An independent human audit with two raters is future work; the packets and decisions are released.

Appendix C. Reproducibility

All steps run from the repository with Docker and Python 3.12:

```
docker build -t aai-c-replay:1 sandbox-replay
python scripts/replay_cpack.py --data DATA --split-lock SPLIT \
    --output REPLAY
python scripts/build_mechanism_benchmark.py --data DATA \
    --replay REPLAY --split-lock SPLIT --output BENCH_V1
python scripts/relabel_repairs.py --data DATA --bench BENCH_V1 \
    --audit AUDIT --output BENCH
python scripts/run_mechanism_eval.py --partition validation ...
python scripts/run_mechanism_eval.py --partition test \
    --scopes partition full --penalty 1.0 ...
```

Every run records the code fingerprint at start, the container image, input and split hashes. The figures are redrawn from the committed run artifacts with `python paper/make_figures.py --lang en`; Figure 1 reads the three programs exported by `paper/export_example.py`, and Figure 8 a screenshot taken by `paper/capture_tool.py` from a fresh instance of the tool. The replay took 14 minutes on a four-core Docker VM; building the benchmarks executed 7,040 delta-debugging probes and 4,350 candidate mutants.

Appendix D. Per-category recovery

Exploratory analysis (amendment A7): share of items whose K-means cluster (oracle k , seed 42) has the item's category as its majority label.

Table D.1: Per-category recovery on real repairs (full cohorts).

Category	n	Signature	Outcomes	Comb.+stdout	Deviation	Evidence	Embedding
output text	177	0.98	0.98	0.92	0.95	0.94	0.92
computation	28	0.54	0.54	0.64	0.57	0.61	0.32
missing statement	27	0.37	0.37	0.67	0.56	0.63	0.70
branch condition	17	0.29	0.24	0.65	0.59	0.59	0.18
loop boundary	16	0.88	0.88	0.94	0.81	0.81	0.31
output format	13	0.00	0.00	0.69	0.69	0.69	0.77
control flow	12	0.83	0.83	0.83	0.83	0.92	0.33
extra statement	8	0.12	0.12	0.38	0.50	0.50	0.62
initialisation	7	0.57	0.57	1.00	1.00	1.00	0.57
input	6	0.50	0.50	0.50	0.50	0.50	0.50
statement placement	6	0.50	0.50	0.33	0.50	0.33	0.33
numeric type	3	1.00	1.00	1.00	1.00	1.00	0.67

Table D.2: Per-category recovery on injected mutants (test cohorts).

Category	n	Signature	Outcomes	Comb.+stdout	Deviation	Evidence	Embedding
missing statement	72	0.56	0.49	0.69	0.60	0.64	0.18
extra statement	65	0.49	0.46	0.68	0.82	0.82	0.57
output text	60	0.62	0.62	0.90	0.97	0.95	0.18
initialisation	53	0.40	0.40	0.60	0.53	0.58	0.15
loop boundary	50	0.62	0.56	0.68	0.62	0.62	0.20
statement placement	48	0.12	0.08	0.67	0.79	0.58	0.23
computation	43	0.51	0.51	0.70	0.67	0.77	0.79
branch condition	29	0.38	0.34	0.45	0.45	0.48	0.93
output format	9	1.00	1.00	1.00	1.00	0.89	0.00

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